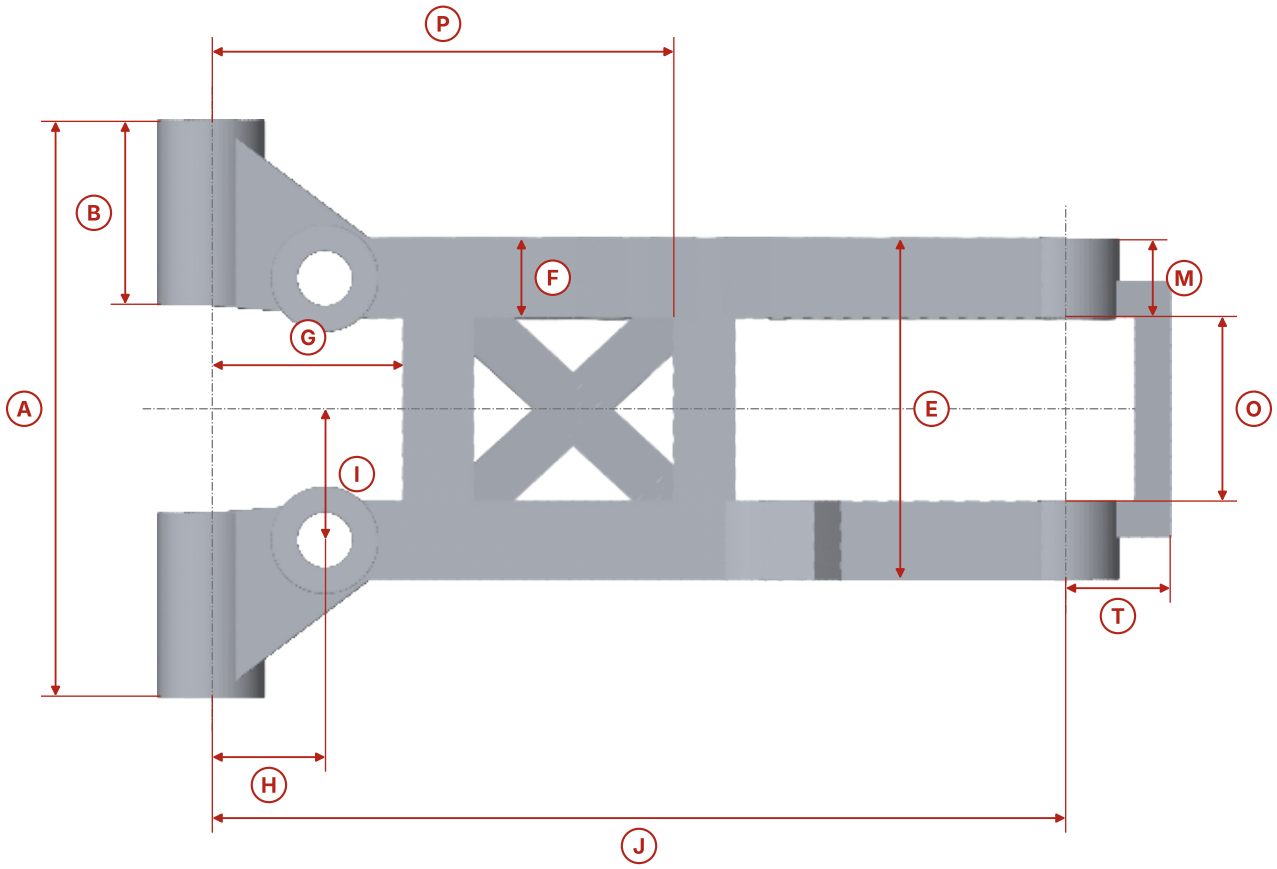
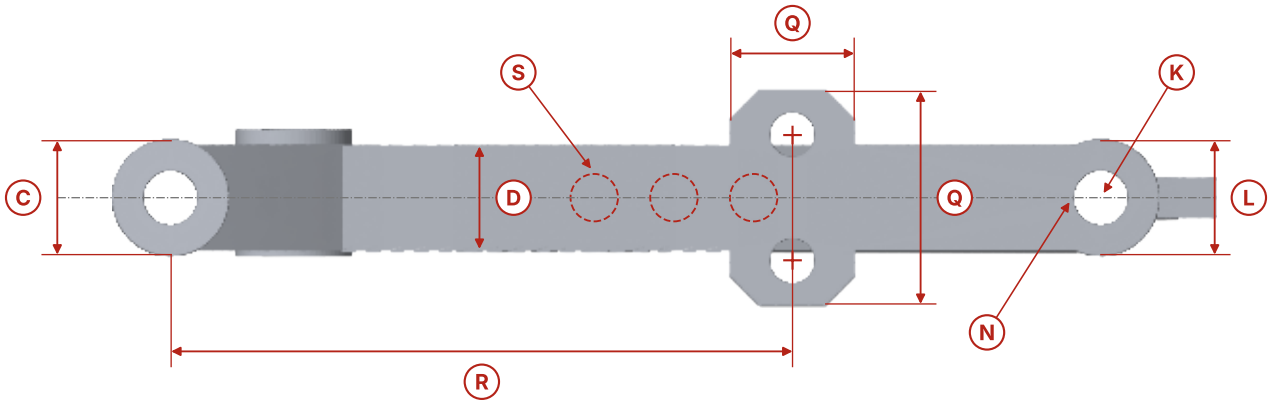
Measurement sheet: Front wishbone (End A, 2 pieces)

Stock part of the Carten T410R (NHA402), not part of the speedrun design. Dimensions in mm. The drawing is the current model (estimate from the photos).



Plan view from above. Rail A (with the octagonal shock plate) at the bottom, rail B (sway bar holes) at the top.
Inner hinge pin on the left, outer hinge pin on the right (dash-dot lines: pin axes and middle of the arm).

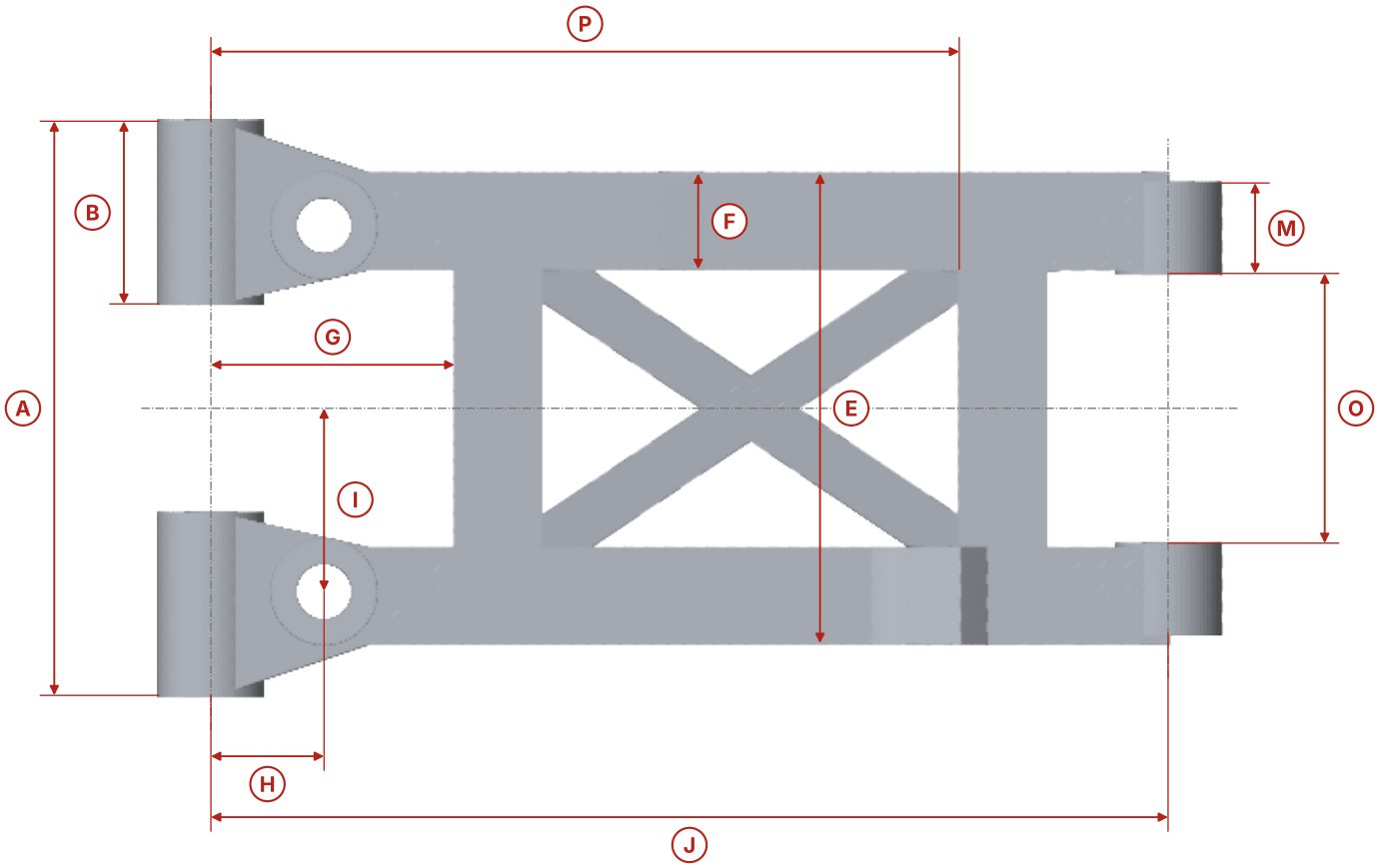


Side view of rail A. R: holes of the shock ball stud (+). S: holes of the sway bar ball stud in rail B (dashed, behind).
K: diameter of the outer hinge pin. N: height of the outer pin axis above the inner pin axis (0 = same height).

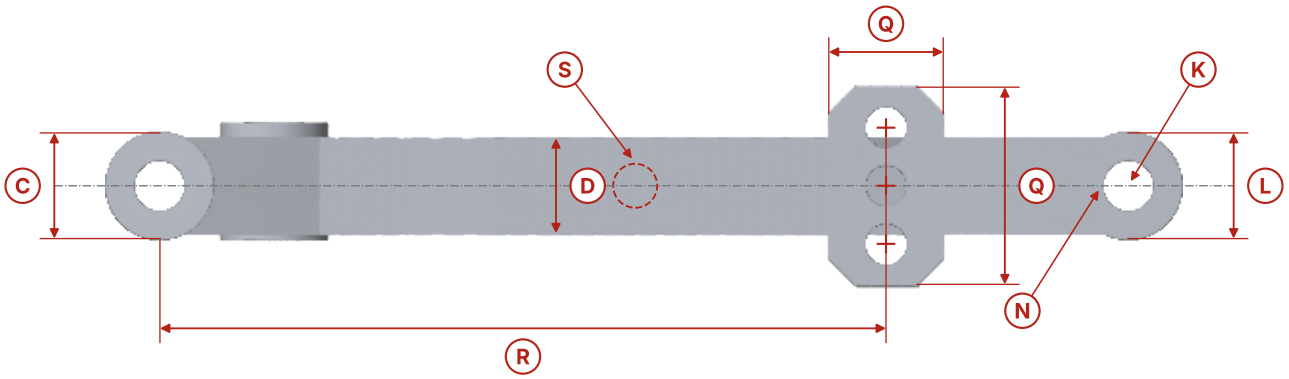
All x values: distance from the axis of the inner hinge pin toward the wheel. All z values: height above the axis of the inner hinge pin.
Hole positions: measure from the edge of the bore of the inner hinge pin to the edge of the hole and add half of both diameters.
The instructions for each letter are on page 3. Write the values in the table on page 3.

Measurement sheet: Rear wishbone (End B, 2 pieces)

Stock part of the Carten T410R (NHA402), not part of the speedrun design. Dimensions in mm. The drawing is the current model (estimate from the photos).



Plan view from above. Rail A (with the octagonal shock plate) at the bottom, rail B (sway bar holes) at the top.
Inner hinge pin on the left, outer hinge pin on the right (dash-dot lines: pin axes and middle of the arm).



Side view of rail A. R: holes of the shock ball stud (+). S: holes of the sway bar ball stud in rail B (dashed, behind).
K: diameter of the outer hinge pin. N: height of the outer pin axis above the inner pin axis (0 = same height).

All x values: distance from the axis of the inner hinge pin toward the wheel. All z values: height above the axis of the inner hinge pin.
Hole positions: measure from the edge of the bore of the inner hinge pin to the edge of the hole and add half of both diameters.
The instructions for each letter are on page 3. Write the values in the table on page 3.

Measurement sheet: wishbones, values

Model: current value in the model (estimate). Measured: write the caliper value. Dimensions in mm.

Letter	What to measure	Front: model	Front: measured	Rear: model	Rear: measured
A	Length along the inner hinge pin over the outer faces of the two inner lugs.	33		33	
B	Length of one inner lug along the hinge pin.	10.5		10.5	
C	Outer diameter of the inner lugs.	6		6	
D	Height of the side rails (top face to bottom face).	5.4		5.4	
E	Width over the outer faces of the two side rails.	19.5		27	
F	Width of one side rail.	4.45		5.5	
G	Inner pin axis to the inner cross bar at the end of the notch between the inner lugs (start).	11		14	
H	Inner pin axis to the axis of the droop screw bosses (M4 thread).	6.5		6.5	
I	Middle of the arm to the axis of a droop screw boss (or: distance of the two axes / 2).	7.5		10.5	
J	Inner pin axis to outer pin axis: outer edges of the two bores minus one bore diameter.	49		55	
K	Diameter of the outer hinge pin (or of the bore of the outer lugs).	3		3	
L	Outer diameter of the outer lugs.	6		6	
M	Length of one outer lug along the outer hinge pin.	4.4		5.2	
N	Arm on a flat surface: height of the outer pin axis minus height of the inner pin axis.	0		0	
O	Gap between the two outer lugs.	10.6		15.5	
P	Inner pin axis to the outer cross bar (start of the cross bar).	26.5		43	
Q	Octagonal shock plate: inner pin axis to its start and to its end (x); how far it reaches above and below the inner pin axis (z).	x 29.5 / 36; z -5.6 / 5.6		x 38 / 44.5; z -5.6 / 5.6	
R	Each hole of the shock ball stud: x and z of its centre.	(32.75 / 3.3) (32.75 / -3.3)		(41.25 / 3.3) (41.25 / 0) (41.25 / -3.3)	
S	Each hole of the sway bar ball stud in rail B: x and z of its centre.	(22.3 / 0) (26.5 / 0) (30.7 / 0)		(27 / 0)	
T	Front arm only: outer pin axis to the outer edge of the thin loop beyond the outer lugs; width of the loop.	6; width 14.6		-	

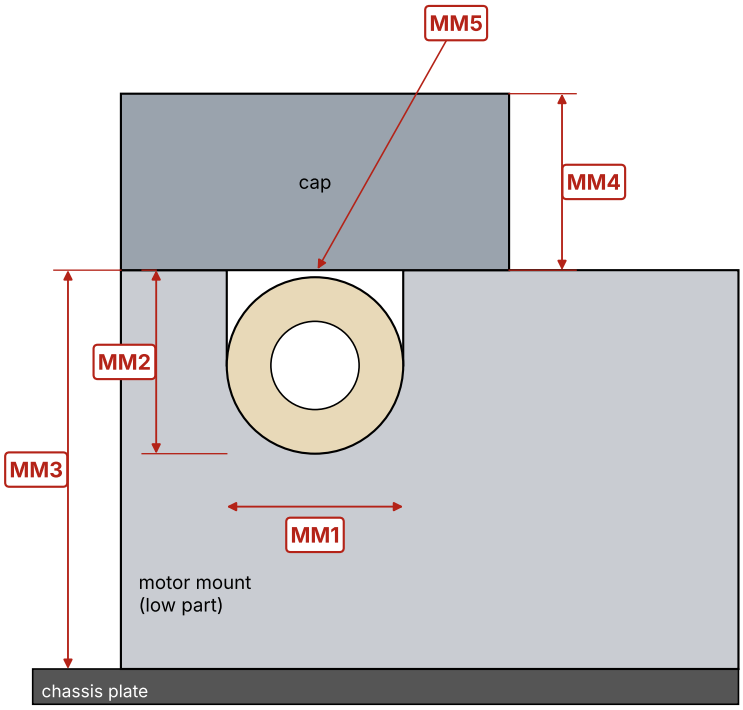
Also please answer: does the octagonal shock plate reach equally far above and below the side rail? (Q: write both values.)
If not everything can be measured: J, A, O and Q are the most important values for the fit on the car.

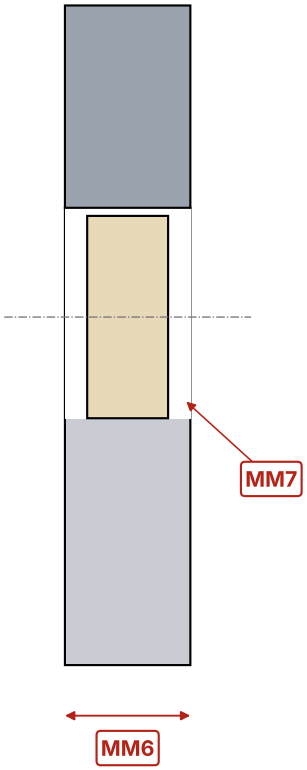
Measurement sheet: bearing seat of the motor mount (NBA201)

Stock part, not part of the speedrun design. Bearing NBA298 5×10×4 (manual step 5). Dimensions in mm.

Section along the shaft

View along the pinion shaft





No.	What to measure	Model	Measured
MM1	Width of the U notch (bearing seat).	10	
MM2	Depth of the U notch: top face of the low part to the bottom of the U.	10.4	
MM3	Height of the low part: bottom face of the motor mount to its top face (below the cap).	22.6	
MM4	Height of the cap.	10	
MM5	Does the cap touch the bearing? If not: gap between the bearing and the cap.	0.4 gap	
MM6	Thickness of the motor mount at the bearing seat (along the shaft).	6.2	
MM7	Is there a step (shoulder) in the seat beside the bearing? On which side: spur gear or cup joint? Diameter of the smaller bore and width of the seat.	no step	
MM8	Where does the set screw M3×3 of the cap go, and what does it press on?	-	